

HDOC - Day 22

Menu-Driven Operations Using Nested Loops

Aim

To write a menu-driven C program that accepts two positive integers m and n ($m \leq n$) and uses a switch statement to display multiplication tables, a multiplication grid, consecutive integers, or exit. The menu repeats until Exit is selected and invalid choices are handled.

Algorithm

1. Start.
2. Read positive integers m and n. Repeat input until $m \leq n$.
3. Display the four-option menu and read choice.
4. Choice 1: using a loop from 1 to 10, print the multiplication tables of m and n side by side.
5. Choice 2: use nested loops to print the multiplication grid for rows 1 to m and columns 1 to n.
6. Choice 3: for every integer i from m to n, print i followed by the next five consecutive integers.
7. Choice 4: exit. For any other choice, display an error message.
8. Repeat the menu until choice 4 is selected. Stop.

Prepared Test Case Table

Test	m,n	Choice	Operation	Expected result
1	3,5	1	Tables	Tables of 3 and 5, multipliers 1-10
2	3,5	2	Grid	3 x 5 multiplication grid
3	3,5	3	Consecutive	Rows 3 to 5, each with next five integers
4	3,5	7 then 4	Invalid + Exit	Error message, menu repeats, then exits

C Program

```
#include <stdio.h>

int main() {
    int m, n, choice, i, j;

    do {
        printf("Enter m and n ( $m \leq n$ ): ");
        scanf("%d %d", &m, &n);
        if(m <= 0 || n <= 0 || m > n)
            printf("Invalid input. Enter positive integers with  $m \leq n$ .\n");
    } while(m <= 0 || n <= 0 || m > n);

    do {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\nEnter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                printf("Multiplication Tables of %d and %d\n", m, n);
                for(i=1; i<=10; i++)
                    printf("%d x %d = %-4d    %d x %d = %d\n",
                        m,i,m*i,n,i,n*i);
                break;
            case 2:
                printf("Multiplication Grid (m=%d, n=%d)\n", m,n);
                for(j=1; j<=n; j++) printf("%4d",j);
                printf("\n    ");
                for(j=1; j<=n; j++) printf("----");
                printf("\n");
                for(i=1; i<=m; i++) {
                    printf("%d | ",i);
                    for(j=1; j<=n; j++) printf("%4d",i*j);
                    printf("\n");
                }
                break;
            case 3:
                printf("Consecutive Integers from %d to %d\n",m,n);
                for(i=m; i<=n; i++) {
                    printf("%d: ",i);
                    for(j=i; j<=i+5; j++) printf("%d ",j);
                    printf("\n");
                }
                break;
            case 4:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid menu choice. Please try again.\n");
        }
    }
    while(choice != 4);
    return 0;
}
```

}

Compilation, Execution & Testing

Test Case 1 - Multiplication Tables

```
● ● ●
puppie@Anvikas-MacBook-Air Desktop % ./hdoc22
Enter m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1
Multiplication Tables of 3 and 5
3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 2 - Multiplication Grid

```
● ● ●
puppie@Anvikas-MacBook-Air Desktop % ./hdoc22
Enter m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2
Multiplication Grid (m=3, n=5)
  1  2  3  4  5
-----
1 |  1  2  3  4  5
2 |  2  4  6  8 10
3 |  3  6  9 12 15
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 3 - Consecutive Integers

```
● ● ●
puppie@Anvikas-MacBook-Air Desktop % ./hdoc22
Enter m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3
Consecutive Integers from 3 to 5
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 4 - Invalid Choice and Exit

```
puppie@Anvikas-MacBook-Air Desktop % ./hdoc22
Enter m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 7
Invalid menu choice. Please try again.

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4
Exiting program.
puppie@Anvikas-MacBook-Air Desktop %
```

Result

The menu-driven C program was successfully implemented and tested for multiplication tables, multiplication grid, consecutive integers, invalid menu choice handling and exit.